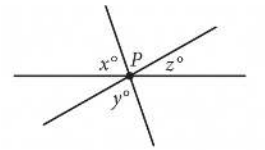


Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.

In the figure, three lines intersect at point P. If $x = 65$ and $y = 75$, what is the value of z ?

Answer

- A. 140
- B. 80
- C. 40
- D. 20

Correct Answer: C

Rationale

Choice C is correct. The angle that is shown as lying between the y° angle and the z° angle is a vertical angle with the x° angle. Since vertical angles are congruent and $x = 65$, the angle between the y° angle and the z° angle measures 65° . Since the 65° angle, the y° angle, and the z° angle are adjacent and form a straight angle, it follows that the sum of the measures of these three angles is 180° , which is represented by the equation $65^\circ + y^\circ + z^\circ = 180^\circ$. It's given that $y = 75$. Substituting 75 for y yields $65^\circ + 75^\circ + z^\circ = 180^\circ$, which can be rewritten as $140^\circ + z^\circ = 180^\circ$. Subtracting 140° from both sides of this equation yields $z^\circ = 40^\circ$. Therefore, $z = 40$.

Choice A is incorrect and may result from finding the value of $x + y$ rather than z . Choices B and D are incorrect and may result from conceptual or computational errors.